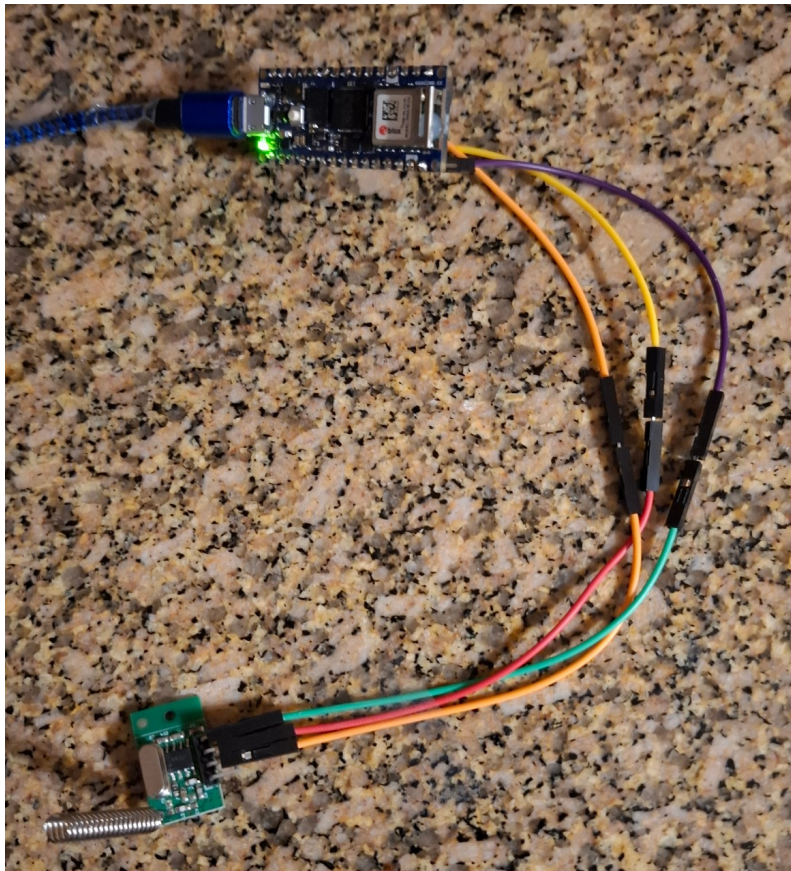


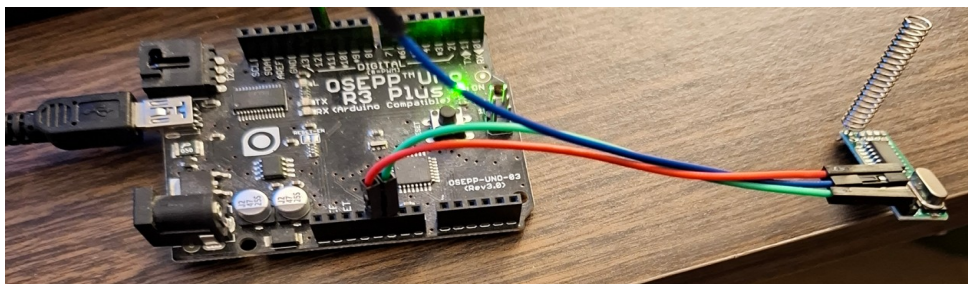
XX

This the Arduino Nano RP2040 connected to the 433 MHz transmitter.

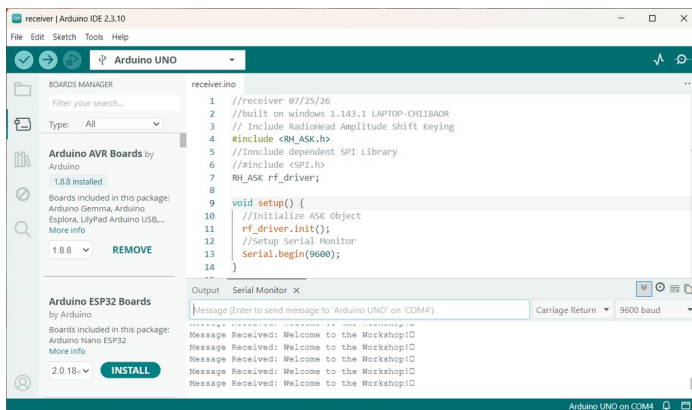


tx

This is the Arduino Uno connected to the 433 MHz receiver.



rx



tx

Installed sudo apt update and sudo apt install arduino
sudo /usr/bin/arduino

devel@pi4-50:~\$ dpkg -l | sort | grep arduino

ii arduino 2:1.8.19+dfsg1-1

IDE from Arduino CC

ii arduino-builder 1.3.25-4+b3

compiling Arduino sketches

ii arduino-core-avr 1.8.7+dfsg-1~deb12u1

AVR microcontroller

ii arduino-ctags 5.8-arduino11-2

exuberant ctags

arduino nano rp2040

arm64 AVR development board

arm64 Command line tool for

all Arduino Core for

arm64 Arduino fork of

author=Mike McCauley <mikem@airspayce.com>
maintainer=Mike McCauley <mikem@airspayce.com>
sentence= Packet Radio library for embedded microprocessors
paragraph=Provides a complete object-oriented library for sending and receiving packetized messages via a variety of common data radios and other transports on a range of embedded microprocessors.
category=Communication
url=https://www.airspayce.com/mikem/arduino/RadioHead/
architectures=*
includes=RadioHead.h
depends=
library.properties

root@pi4-50:~/Arduino#/receiver/
receiver.ino

root@pi4-50:~/Arduino/transmitter# less transmitter.ino
//building on pi4-50 Arduino Nano RP2040
//07/25/26 Include RadioHead Amplitude Shift Keying version=1.145
//receiver 07/24/26
// Include RadioHead Amplitude Shift Keying
#include <RH_ASK.h>
//Innclude dependent SPI Library
//#include <SPI.h>
RH_ASK rf_driver;

```
void setup() {  
  //Initialize ASK Object  
  rf_driver.init();  
  //Setup Serial Monitor  
  Serial.begin(9600);  
}
```

```
void loop() {  
  uint8_t buf[24];  
  uint8_t buflen = sizeof(buf);  
  if (rf_driver.recv(buf, &buflen))  
  {  
    Serial.print("Message Received: ");  
    Serial.println((char*)buf);  
  }  
}
```

receiver.ino

transmitter.ino
//building on pi4-50 Arduino Nano RP2040
//07/25/26 Include RadioHead Amplitude Shift Keying version=1.145
#include <RH_ASK.h>
//Innclude dependent SPI Library
#include <SPI.h>
RH_ASK rf_driver;

```
void setup() {
  // put your setup code here, to run once:
  rf_driver.init();
}
```

```
void loop() {
  // put your main code here, to run repeatedly:
  char *msg = "Welcome to the Workshop!";
  rf_driver.send((uint8_t *)msg, strlen(msg));
  rf_driver.waitPacketSent();
}
transmitter.ino
```

devel@LAPTOP-CH1I8AOR:/mnt/c/Users/vidal/Documents/Arduino/libraries/RadioHead\$

```
name=RadioHead
version=1.143.1
author=Mike McCauley <mikem@airspayce.com>
maintainer=Mike McCauley <mikem@airspayce.com>
sentence= Packet Radio library for embedded microprocessors
paragraph=Provides a complete object-oriented library for sending and receiving packetized
messages via a variety of common data radios and other transports on a range of embedded
microprocessors.
category=Communication
url=https://www.airspayce.com/mikem/arduino/RadioHead/
architectures=*
includes=RadioHead.h
depends=
```

devel@LAPTOP-CH1I8AOR:/mnt/c/Users/vidal/Documents/Arduino\$ less
transmitter/transmitter.ino

```
//07/25/26
//built on windows 1.143.1 LAPTOP-CH1I8AOR
//Include RadioHead Amplitude Shift Keying
#include <RH_ASK.h>
//Innclude dependent SPI Library
#include <SPI.h>
RH_ASK rf_driver;
```

```
void setup() {
  // put your setup code here, to run once:
  rf_driver.init();
}
```

```
void loop() {
  // put your main code here, to run repeatedly:
  char *msg = "Welcome to the Workshop!";
  rf_driver.send((uint8_t *)msg, strlen(msg));
  rf_driver.waitPacketSent();
  delay(1000);
}
```

devel@LAPTOP-CH1I8AOR:/mnt/c/Users/vidal/Documents/Arduino\$ less receiver/receiver.ino

```

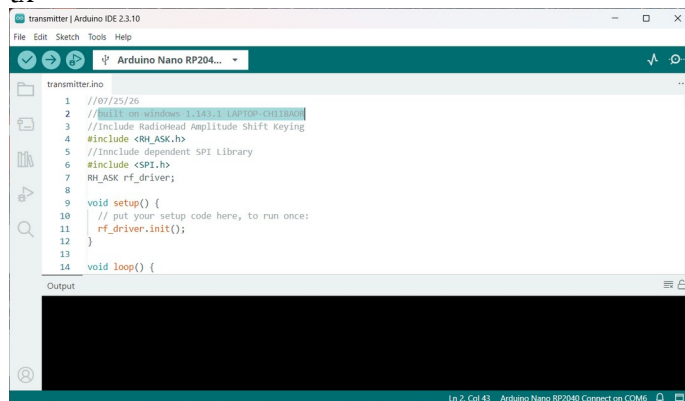
//receiver 07/25/26
//built on windows 1.143.1 LAPTOP-CH1I8AOR
// Include RadioHead Amplitude Shift Keying
#include <RH_ASK.h>
//Innclude dependent SPI Library
//#include <SPI.h>
RH_ASK rf_driver;

void setup() {
  //Initialize ASK Object
  rf_driver.init();
  //Setup Serial Monitor
  Serial.begin(9600);
}

void loop() {
  uint8_t buf[24];
  uint8_t buflen = sizeof(buf);
  if (rf_driver.recv(buf, &buflen))
  {
    Serial.print("Message Received: ");
    Serial.println((char*)buf);
  }
}

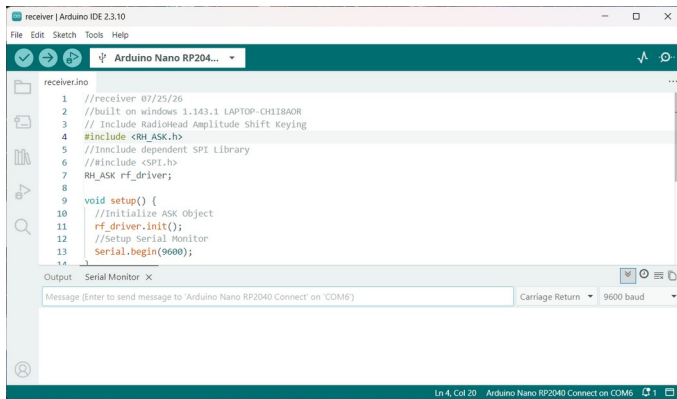
```

tx



tx

rx



TX

could not fork this repo <https://github.com/hallard/RadioHead.git>
 now included in <https://github.com/develone/raspi-ask-433-tests>

To use the **RadioHead** ASK (Amplitude Shift Keying) library for a 433/315 MHz receiver on a Raspberry Pi in C, you must rely on the wrapper for the underlying C++ library and the bcm2835 hardware driver. However, due to the Pi's multitasking OS, it cannot handle the strict, cycle-accurate bit-banging required for basic ASK, so you must use hardware-accelerated modules like the **RFM69** or **RFM95**

Because the standard ASK classes in the [RadioHead Packet Radio Library](#) were originally designed for microcontrollers with dedicated timers, using cheap ASK receivers (like the FS1000A) natively on the Raspberry Pi often results in poor performance. Instead, connecting an Arduino to your Pi via USB/Serial to do the radio heavy-lifting is the community's preferred workaround

Setting Up the Environment (RFM Modules)

If you are using a supported module (e.g., RFM69) wired to the Pi's SPI pins, you will need to compile the C++ example using the bcm2835 library

```

cd bcm2835-1.75
./configure make
sudo make check
sudo make install

```

Repo <https://github.com/develone/raspi-ask-433-tests>

```

transmitter
sudo apt update
sudo apt-get install pigpio

```

```
gcc -Wall -pthread -o ask_transmitter ask_transmitter.c -lpigpio -lrt
```

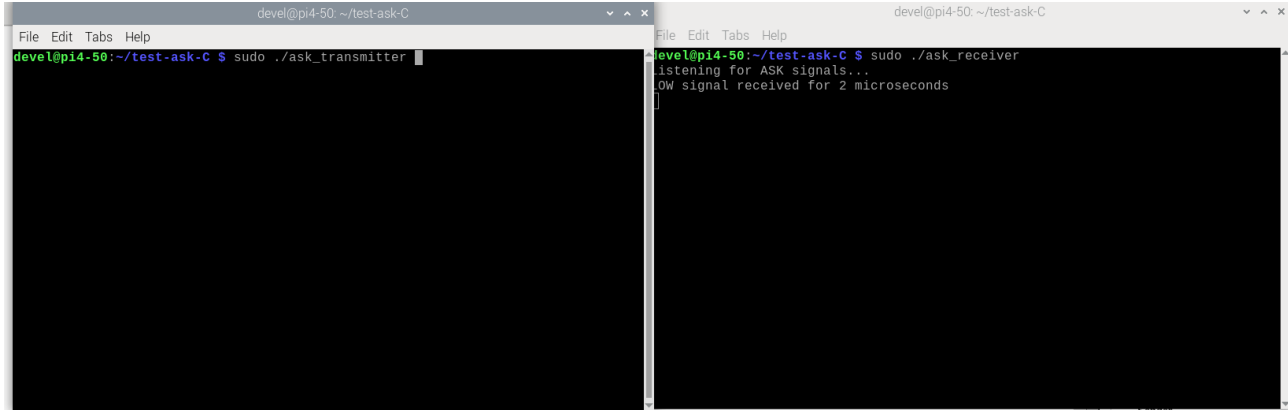
```
devel@pi4-50:~/test-ask-C $ sudo ./ask_transmitter
```

receiver

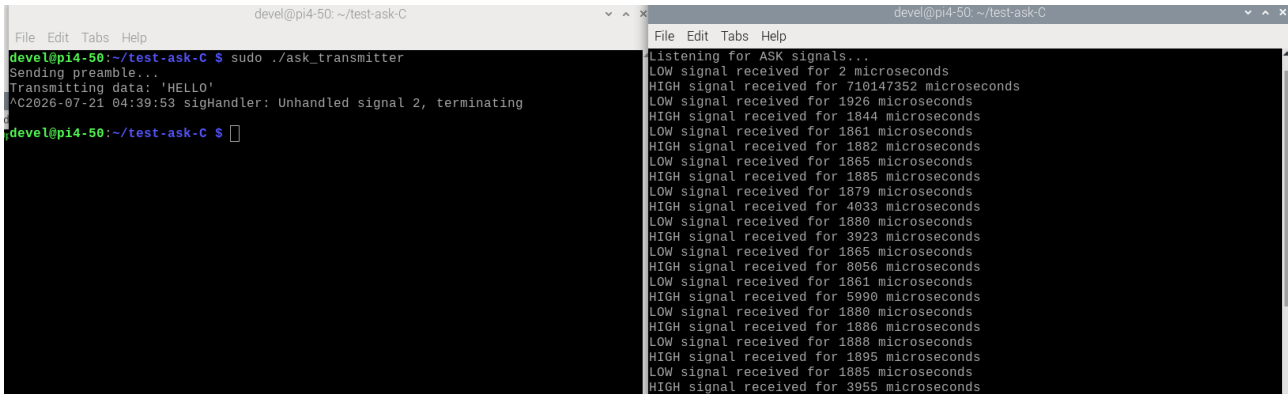
Requires WiringPi <https://github.com/WiringPi/WiringPi.git>
 git clone <https://github.com/WiringPi/WiringPi.git>
 cd WiringPi
 ./build

gcc -o ask_receiver ask_receiver.c -lwiringPi
 devel@pi4-50:~/test-ask-C \$ sudo ./ask_receiver

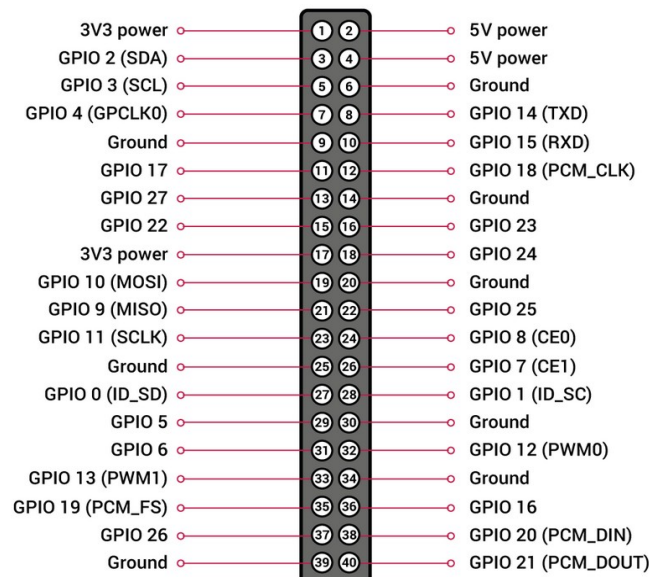
transmitter



receiver



pi-connector



Needs number of char sent

Needs a CRC